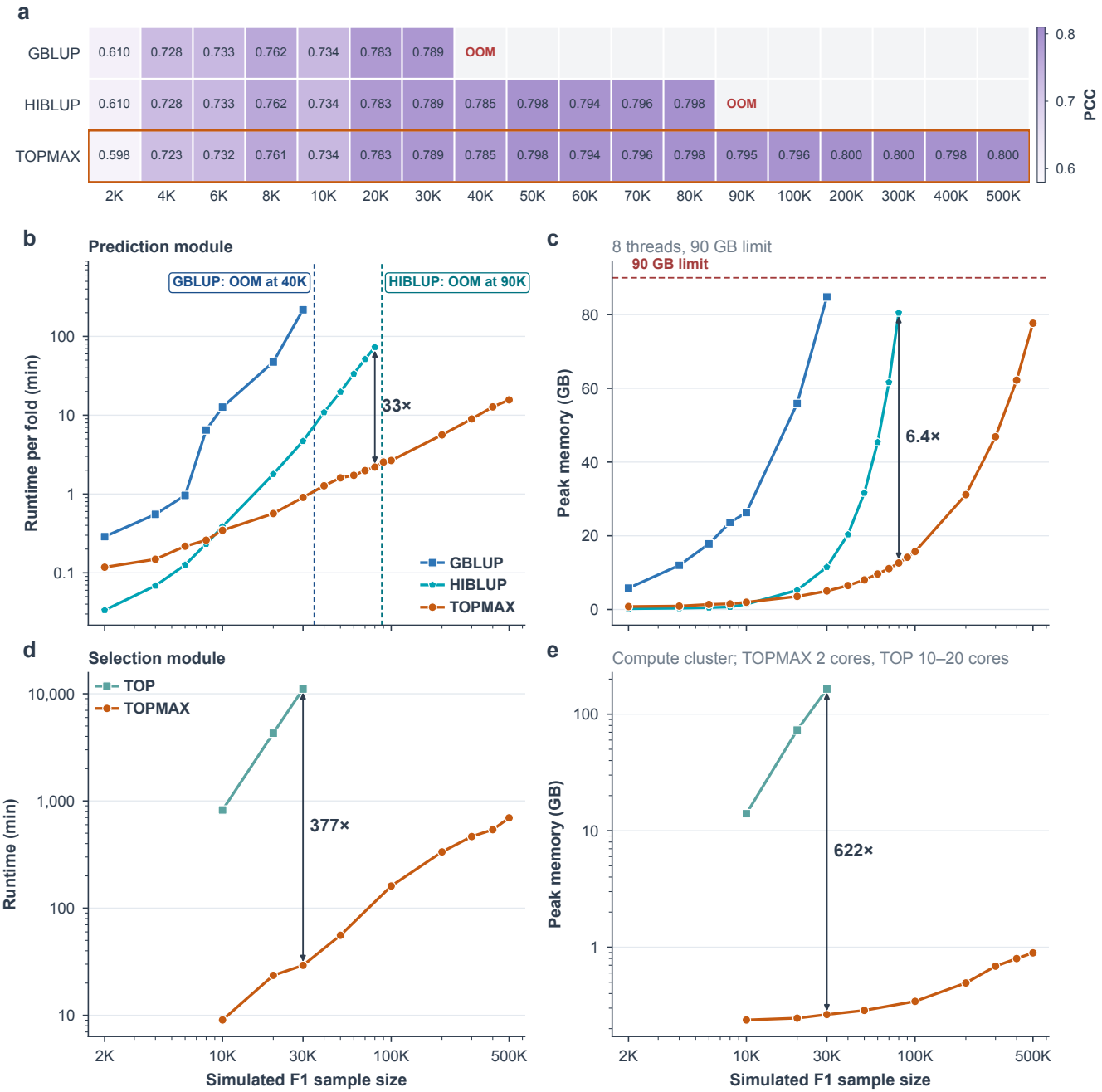

a
GBLUP
HIBLUP
TOPMAX
0.610 0.728 0.733 0.762 0.734 0.783 0.789 OOM
0.610 0.728 0.733 0.762 0.734 0.783 0.789 0.785 0.798 0.794 0.796 0.798 OOM
0.598 0.723 0.732 0.761 0.734 0.783 0.789 0.785 0.798 0.794 0.796 0.798 0.795 0.796 0.800 0.800 0.798 0.800
2K 4K 6K 8K 10K 20K 30K 40K 50K 60K 70K 80K 90K 100K 200K 300K 400K 500K
PCC
0.8
0.7
0.6
b
Prediction module
GBLUP: OOM at 40K
HIBLUP: OOM at 90K
Runtime per fold (min)
100
10
1
0.1
33×
GBLUP
HIBLUP
TOPMAX
c
8 threads, 90 GB limit
90 GB limit
Peak memory (GB)
80
60
40
20
0
6.4×
d
Selection module
TOP
TOPMAX
Runtime (min)
10,000
1,000
100
10
377×
2K
10K
30K
100K
500K
Simulated F1 sample size
e
Compute cluster; TOPMAX 2 cores, TOP 10–20 cores
Peak memory (GB)
100
10
1
622×
2K
10K
30K
100K
500K
Simulated F1 sample size